National University of Computer and Emerging Sciences

FAST LAHORE FALL 2022

Assignment # 3 **Applied Physics** **Max. Marks: 50**

Section: BCS 1B **Submission Date: 24th October, 2022 (Monday)**

Instructions:

1. ***First page of the assignment should be the page given below***
2. ***Assignment should be hand-written. Typed assignment will not be accepted.***
3. ***Use A4 Paper for the assignment. Do not use any other sheet***
4. ***Please write the answers in the space given below.***
5. ***Make diagrams wherever possible.***

ROLL NO.: - -------------------------NAME: --

Question #1: A 200-g block connected to a light spring for which the force constant is 5.00 N/m is free to oscillate on a frictionless, horizontal surface. The block is displaced 5.00 cm from equilibrium and released from rest. Find the period of its motion.

(Marks 10)

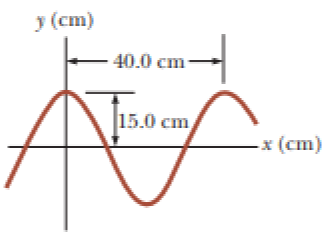
Question #2: Mass of 1300 kg is constructed so that its frame is supported by four springs. Each spring has a force constant of 20,000 N/m. Two people riding in the car have a combined mass of 160 kg. Find the frequency of vibration of the car after it is driven over a pothole in the road.

(Marks 10)

Question #3: A sinusoidal wave traveling in the positive x direction has an amplitude of 15.0 cm, a wavelength of 40.0 cm, and a frequency of 8.00 Hz. The vertical position of an element of the medium at t=0 and x=0 is also 15.0 cm as shown in Figure.

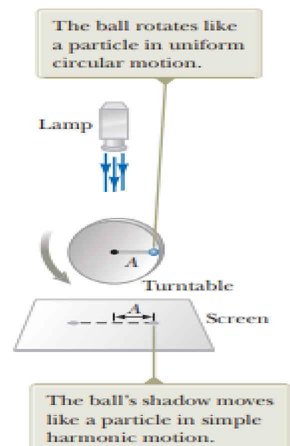
(A) Find the wave number k, period T, angular frequency v, and speed v of the wave.

(Marks 10)



Question #4: The ball in Figure rotates counterclockwise in a circle of radius 3.00 m with a constant angular speed of 8.00 rad/s. At t= 0, its shadow has an x coordinate of 2.00 m and is moving to the right. Determine the x coordinate of the shadow as a function of time in SI units.

(Marks 10)



<u>Question #5:</u> A pulse moving to the right along the x axis is represented by the wave function here x and y are measured in centimeters and t is measured in seconds. Find expressions for the wave function at t=0, t=1.0 s, and t=2.0 s.

$$y(x, t) = \frac{2}{(x - 3.0t)^2 + 1}$$

(Marks 10)